The Journal of Open Source Software

DSSE: An environment for simulation of reinforcement learning-empowered drone swarm maritime search and rescue missions

Renato Laffranchi Falcão [1*¶], **Jorás Custódio Campos de Oliveira** [1*], **Pedro Henrique Britto Aragão Andrade** [1*], **Ricardo Ribeiro Rodrigues** [1*], **Fabrício Jailson Barth** [1*], and **José Fernando Basso Brancalion** [2*]

1 Insper, Brazil **2** Embraer, Brazil ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Kanishka B. Narayan

Reviewers:

- @draabe
- @wob86

Submitted: 02 May 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The goal of this project is to contribute to the research of solutions that employ reinforcement learning techniques to maritime search and rescue missions of shipwrecked people. The software equip's researchers with a simulation of shipwrecked people casted away according to maritime currents to producing a stochastic environment to be used to train and evaluate autonomous agents.

Statement of need

Maritime navigation plays a crucial role across various domains, including leisure activities and commercial fishing. However, maritime transportation is particularly significant as it accounts for 80% to 90% of global trade (*Safety and Shipping Review*, 2023). While maritime navigation is essential for global trade, it also poses significant safety risks, as evidenced by the World Health Organization's report (*Drowning*, 2023) of approximately 236,000 annual drowning deaths worldwide. Therefore, maritime safety is essential, demanding significant enhancements in search and rescue (SAR) missions. It is imperative that SAR missions minimize the search area and maximize the chances of locating the search object.

To achieve this objective, traditional SAR operations used path planning algorithms such as parallel sweep, expanding square, and sector searches (*International Aeronautical and Maritime Search and Rescue Manual - Volume II - Mission Co-Ordination*, 2022). But recent researches propose a different approach to this problem using reinforcement learning techniques over pre-determined search patterns (Ai et al., 2021; Wu et al., 2024).

In order to contribute to researches on the effectiveness of integrating reinforcement learning techniques into SAR path planning, the Drone Swarm Search Environment (DSSE), distributed as a Python package, was designed to provide a training environment using the PettingZoo (J. Terry et al., 2021) interface. Its purpose is to facilitate the training and evaluation of single or multi-agent reinforcement learning algorithms. Additionally, it has been included as a third-party environment in the official PettingZoo documentation (Jordan Terry et al., 2021).

Simulation environment showcasing the algorithm's execution.

Figure 1: Simulation environment showcasing the algorithm's execution.

The environment depicted in Figure 1 comprises a grid, a probability matrix, drones, and an arbitrary number of persons-in-water (PIW). The movement of the PIW is influenced by,

but not identical to, the dynamics of the probability matrix, which models the drift of sea currents impacting the PIW (Wu et al., 2023). The probability matrix itself is defined using a two-dimensional Gaussian distribution, which expands over time, thus broadening the potential search area. This expansion simulates the diffusion of the PIW, approximating the zone where drones are most likely to detect them. Moreover, the environment employs a reward function that incentivizes the speed of the search, rewarding the agents for shorter successful search durations.

Using this environment, any researcher or practitioner can write code and execute an agent's training, such as the source code presented below.

```
from DSSE import DroneSwarmSearch

env = DroneSwarmSearch()

observations, info = env.reset()
rewards = 0
done = False
while not done:
      actions = policy(observations, env.get_agents())
      observations, rewards, terminations, truncations, infos = env.step(actions)
      done = any(terminations.values()) or any(truncations.values())
```

The package also includes a second environment option. Similar to the first, this alternative setup is designed for training agents, but with key differences in its objectives and mechanics. Unlike the first environment, which rewards agents for speed in their searches, this second option rewards agents that cover the largest area without repetition. It incorporates a trade-off by using a stationary probability matrix, but enhances the simulation with a more advanced Lagrangian particle model (Dagestad et al., 2018) for pinpointing the PIW's position. Moreover, this environment omits the inclusion of shipwrecked individuals, focusing instead on promoting research into how agents can learn to efficiently expand their search coverage over broader areas.

The grid is divided into square cells, each representing a quadrant with sides measuring 130 meters in the real world. This correlation with real-world dimensions is crucial for developing agents capable of learning from realistic motion patterns. The drones, which are controlled by reinforcement learning algorithms, serve as these agents. During the environment's instantiation, users define the drones' nominal speeds. These drones can move both orthogonally and diagonally across the grid, and they are equipped to search each cell for the presence of the PIW.

Several works have been developed over the past few years to define better algorithms for the search and rescue of shipwrecks (Ai et al., 2021; Wu et al., 2024). However, no environment for agent training is made available publicly. For this reason, the development and provision of this environment as a Python library and open-source project are expected to have significant relevance to the machine learning community and ocean safety.

This new library makes it possible to implement and evaluate new reinforcement learning algorithms, such as Deep Q-Networks (DQN) (Mnih et al., 2015) and Proximal Policy Optimization (PPO) (Schulman et al., 2017), with little effort. An earlier iteration of this software was utilized in research that compared the Reinforce algorithm with the parallel sweep path planning algorithm (Abreu et al., 2023).

References

Abreu, L. D. M. de, Carrete, L. F. S., Castanares, M., Damiani, E. F., Brancalion, J. F. B., & Barth, F. J. (2023). *Exploration and rescue of shipwreck survivors using reinforcement*

learning-empowered drone swarms (pp. 64–69). Simpósio de Aplicações Operacionais em Áreas de Defesa (SIGE).

Ai, B., Jia, M., Xu, H., Xu, J., Wen, Z., Li, B., & Zhang, D. (2021). Coverage path planning for maritime search and rescue using reinforcement learning. *Ocean Engineering*, *241*, 110098. https://doi.org/10.1016/j.oceaneng.2021.110098

Dagestad, K.-F., Röhrs, J., Breivik, Ø., & Ådlandsvik, B. (2018). OpenDrift v1.0: A generic framework for trajectory modelling. *Geoscientific Model Development*, *11*(4), 1405–1420. https://doi.org/10.5194/gmd-11-1405-2018

Drowning. (2023). World Health Organization. https://www.who.int/news-room/fact-sheets/detail/drowning

International aeronautical and maritime search and rescue manual - volume II - mission co-ordination: Vol. II (9th ed.). (2022). (9th ed.) [Computer software]. International Maritime Organization; International Civil Aviation Organization. https://store.icao.int/en/international-aeronautical-and-maritime-search-and-rescue-manual-volume-ii-mission-co-ordination-d

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, *518*(7540), 529–533. https://doi.org/10.1038/nature14236

Safety and shipping review (p. 4). (2023). Allianz Global Corporate & Specialty. https://commercial.allianz.com/news-and-insights/reports/shipping-safety.html

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). *Proximal policy optimization algorithms*. https://doi.org/10.48550/arXiv.1707.06347

Terry, J., Black, B., Grammel, N., Jayakumar, M., Hari, A., Sullivan, R., Santos, L. S., Dieffendahl, C., Horsch, C., Perez-Vicente, R., & others. (2021). Pettingzoo: Gym for multi-agent reinforcement learning. *Advances in Neural Information Processing Systems*, *34*, 15032–15043.

Terry, Jordan, Black, B., Grammel, N., Jayakumar, M., Hari, A., Sullivan, R., Santos, L., Perez, R., Horsch, C., Dieffendahl, C., Williams, N., & Lokesh, Y. (2021). *PettingZoo: Gym for multi-agent reinforcement learning*. https://github.com/Farama-Foundation/PettingZoo

Wu, J., Cheng, L., & Chu, S. (2023). Modeling the leeway drift characteristics of persons-in-water at a sea-area scale in the seas of china. *Ocean Engineering*, *270*, 113444. https://doi.org/10.1016/j.oceaneng.2022.113444

Wu, J., Cheng, L., Chu, S., & Song, Y. (2024). An autonomous coverage path planning algorithm for maritime search and rescue of persons-in-water based on deep reinforcement learning. *Ocean Engineering*, *291*, 116403. https://doi.org/10.1016/j.oceaneng.2023.116403

DRAFT